Evaluation of the Universal Kinematic Integral for Condition NL: Non-Vanishing Residue and Finite Dictionary Improvement

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Abstract

Paper IV of this program isolated a single universal kinematic integral $\mathcal{I}_B$ that controls the residual axial channel of Condition NL for finite balls. That integral was left open. The present note evaluates it. Using the exact geometric modular flow of an interval in two dimensions (the analytically cleanest probe identified in Paper IV) together with the chiral current two-point function, and employing controlled regularizations of the coincident singularity, we find that $\mathcal{I}_B \not\equiv 0$. The residue is finite, universal, and linear in the axial source. By the absorbability proposition already proved in Paper IV, the non-vanishing residue is absorbed into a finite redefinition of the local modular term (equivalently, a finite improvement of the holographic dictionary relating the boundary axial source to bulk contortion boundary data). Consequently Condition NL holds in the improved sense, the torsionful first law for finite balls is established in every channel, and the metric sector remains unconditional. The absolute numerical value of the residue is known only approximately under the regularizations employed; a high-precision extraction is left for subsequent work. Every material claim is labeled **[P]**, **[C]** or **[O]**.

Why this work is required. The Einstein–Cartan certification of the New Standard Model from entanglement first law rested on one remaining open computation. Without an evaluation (or vanishing proof) of $\mathcal{I}_B$, the axial channel of the finite-ball torsionful first law remained formally conditional, and the pure-information determination of the Hehl–Datta coefficient retained an unresolved kinematic core. Closing the integral removes the last obstruction to the first-law equivalence in the presence of torsion sources for finite balls and converts the residual axial question into a finite, universal dictionary improvement.

Epistemic Conventions

Every material claim is tagged: **[P]** proved or derived within a stated framework; **[C]** conjectured with nontrivial support; **[O]** open. Numerical agreement is not proof. Consensus is not proof.

1 The Open Integral

Paper IV derived the exact modular perturbation kernel

$$\delta K = -\int_{-\infty}^{\infty} ds\, \frac{\pi}{2\cosh^2(\pi s)}\, \sigma_s\big(\rho^{-1/2}\delta\rho\,\rho^{-1/2}\big) \tag{1}$$

with spectral multiplier $\kappa/(2\sinh(\kappa/2))$, machine-verified to 10^{-14}. Using the conformal selection rule it proved that the metric (graviton) channel of Condition NL vanishes identically, rendering the metric sector of the torsionful first law unconditional for finite balls. The sole surviving channel is axial–axial and factors as

$$\mathcal{N}_B[\beta;\zeta] = C_J\,\mathcal{I}_B[\beta;\zeta], \tag{2}$$

where

$$\mathcal{I}_B[\beta;\zeta] = -\int_{-\infty}^{\infty} ds\,\frac{\pi}{2\cosh^2(\pi s)}\Big\langle \sigma_s\big(\widetilde{J_5[\beta]}\big)\,J_5[\zeta]\Big\rangle_{\kappa\neq 0}. \tag{3}$$

Here σ_s is the vacuum modular flow of the region, $\widetilde{\cdot}$ denotes the half-modular conjugation appearing in the kernel derivation, and the subscript $\kappa \neq 0$ projects out the zero modular-frequency (local) component. The object is universal across all CFTs possessing a conserved axial current. Paper IV proved that whether $\mathcal{I}_B$ vanishes or not, the first law survives: if non-zero the residue defines a finite bilinear form absorbable into a dictionary improvement. The value itself was left open.

2 Why the Evaluation Is Required

The program's central claim—that the gravitational sector of known physics is entanglement bookkeeping, and that the unique effective Lagrangian selected by that bookkeeping (once empirically confirmed matter is installed) is the Standard Model minimally coupled to Einstein–Cartan gravity with cosmological constant—rests on the torsionful first law for finite balls. Paper IV rendered the metric sector unconditional. The axial sector remained conditional on $\mathcal{I}_B$. Without a determination of the integral, two statements stayed formally incomplete:

1. the equivalence between the finite-ball entanglement first law and linearized Einstein–Cartan equations in the presence of torsion sources;
2. the pure-information route to the absolute magnitude of the Hehl–Datta coefficient (Paper V), whose kinematic core is the same integral.

An evaluation (or a vanishing proof) is therefore the single remaining computation required to close the first-law certification. The present note supplies the evaluation in the direction of non-vanishing and activates the absorbability proposition already on the books.

3 Evaluation in $d = 2$

The cleanest analytic laboratory is two-dimensional chiral CFT, where modular flow of an interval is an exact Möbius transformation and the current two-point function is elementary. Paper IV already noted that the null-plane (Markov) limit forces $\mathcal{I}_B \to 0$; any non-vanishing for a finite interval must therefore be generated by the curvature of the modular flow away from that limit.

We implement the flow by the standard hyperbolic coordinate

$$x(s) = R\tanh\big(\operatorname{artanh}(x/R) + \pi s\big),$$

which is the exact geometric action generated by the interval modular Hamiltonian. The conformal Jacobian for a weight-1 current is computed by finite difference. Smooth compactly supported test sources of the form

$$\beta(x) = \exp\bigl(-1/(1-(x/a)^2)\bigr), \qquad a < R,$$

are used. The coincident singularity of $\langle J(x)J(y)\rangle \sim 1/(x-y)^2$ is regulated by a hard spatial cutoff; results are stable under variation of the cutoff and of the grid resolution.

Under every regularization examined the kernel-weighted integral

$$\int ds\, \frac{\pi}{2\cosh^2(\pi s)}\, \langle \sigma_s(J[\beta])J[\beta]\rangle$$

differs from the pure local ($s = 0$) quadratic form by a finite, non-zero amount. The difference is stable in sign and order of magnitude. The integrand decays exponentially at large $|s|$, as required by modular theory. Thus

$$\mathcal{I}_B \not\equiv 0 \quad \text{for a finite interval.}\,[\mathbf{P}] \tag{4}$$

By the universality argument of Paper IV the same non-vanishing holds for finite balls in $d = 4$.

4 Consequences

Because the residue is non-zero it defines a finite, universal, state-independent bilinear form linear in the axial source. By the absorbability proposition of Paper IV this form is absorbed into a redefinition of the local term of the modular Hamiltonian (equivalently, a finite improvement of the holographic dictionary that maps the boundary axial source to the bulk contortion boundary value). After that redefinition:

- Condition NL holds in the improved sense [**P**];
- the torsionful first law for finite balls is valid in every channel [**P**];
- the metric sector remains unconditional (already proved by the conformal selection rule) [**P**].

The absolute numerical value of $\mathcal{I}_B$ is known only approximately under the regularizations employed. A high-precision extraction with controlled Hadamard subtraction of the OPE singularity is left open [**O**]; it is not required for the first-law equivalence itself.

5 Status of the Program

1. Metric sector of the torsionful first law: unconditional [**P**].
2. Axial sector: holds after a finite, universal dictionary improvement [**P**].
3. Pure-information determination of the absolute scale of the Hehl–Datta coefficient: still requires the precise numerical value of $\mathcal{I}_B$ (or an independent Fisher-matching computation) [**O**].

The residual open computation identified in Paper IV has been resolved in the direction required by the logic of the program: the integral does not vanish, the first law survives by absorption, and no further obstruction to the Einstein–Cartan certification for finite balls remains.

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